

Mirror symmetry for ADE singularities, with applications

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[arXiv:2501.05753](#) — with A. Brini and I. Strachan

[arXiv:2607.07355](#) — with B. Fang and Z. Lan

Remodeling of toric Calabi–Yau threefolds

1. Hori–Vafa mirror for toric Calabi–Yau threefolds

For a toric Calabi–Yau threefold X_Σ , the lattice generators of Σ determine the Newton polynomial

$$H_q(X, Y) = \sum_i a_i X^{m_{i,1}} Y^{m_{i,2}}.$$

The Hori–Vafa mirror is $\check{X}_q = \{uv = H_q(X, Y)\}$; its mirror curve is $\Sigma_q = \{H_q(X, Y) = 0\}$.

2. The mirror curve carries topological-recursion data

$$(\Sigma_q, x, y, B), \quad x = \log X, \quad y = \log Y, \quad \omega_{0,1} = y \, dx, \quad \omega_{0,2} = B,$$

where the bidifferential B is the normalized Bergman kernel.

Topological recursion gives symmetric meromorphic n -differentials $\omega_{g,n}^{\text{TR}}$ on Σ_q^n .

3. Remodeling (Fang-Liu-Zong)

After the closed and open mirror maps,

$$\omega_{g,n}^{\text{TR}} \longleftrightarrow \text{Gromov–Witten generating functions of } X.$$

What replaces the toric mirror curve?

Let $\Gamma \subset SU(2)$ be finite, the corresponding Kleinian singularity is

$$\mathbb{C}^2/\Gamma$$

The McKay correspondence attaches to Γ a simply-laced root system $\mathfrak{g}_\Gamma$. The ADE classification is

$$\mathbb{Z}_{n+1} \longleftrightarrow A_n, \quad \text{binary dihedral} \longleftrightarrow D_n, \quad (2T, 2O, 2I) \longleftrightarrow (E_6, E_7, E_8),$$

where $2T$, $2O$, and $2I$ are the binary tetrahedral, octahedral, and icosahedral groups. Take the minimal resolution $Y_\Gamma \rightarrow \mathbb{C}^2/\Gamma$, they are toric only when Γ is abelian.

Main question

Can we do the same but for the *ADE* singularities and their minimal resolutions?

Two theorems behind the talk

Theorem: ADE Landau–Ginzburg mirrors (Brini–Ma–Strachan)

For every finite subgroup $\Gamma \subset SU(2)$ of ADE type, one can construct

$$P_q^\Gamma(\lambda, \mu) \in \mathbb{Z}[\lambda, \mu],$$

such that solving λ from $P_q^\Gamma(\lambda, \mu) = 0$, $\log \lambda$ is the Landau–Ginzburg mirror of Y_Γ .

Theorem: Binary-dihedral remodeling (Fang–Lan–Ma)

If Γ is a binary dihedral group, let Σ_Γ be the (compactified) zero locus of $P_q^\Gamma(\lambda, \mu)$. Then the mirror of $\mathcal{X} = [\mathbb{C}^2/\Gamma \times \mathbb{C}]$ is described by the $\mathbb{Z}_2$ -equivariant topological recursion associated with (Σ_Γ, x, y, B) .

The exceptional curves form the ADE Dynkin diagram

The exceptional locus of Y_Γ is a configuration of (-2) -curves

$$E_i \cdot E_j = -C_{ij},$$

where C is the Cartan matrix of type A , D , or E .

Classical geometry

- $H_2(Y_\Gamma)$ is the root lattice;
- positive roots encode effective curve classes.

Quantum cohomology

The (small) quantum cohomology is a deformation of the usual cohomology ring by $H_2(Y_\Gamma)$.

Quantum McKay correspondence conjecture

For $\beta = \sum_i b_i \alpha_i \in R_\Gamma^+$, set $q^\beta = \prod_i q_i^{b_i}$, and let t be the equivariant parameter.

Theorem: Bryan–Gholampour quantum product

For $v, w \in H^2(Y_\Gamma)$, their quantum product $v \star w \in H^2(Y_\Gamma)[[q^\beta]]$

$$v \star w = -t^2 |\Gamma| \langle v, w \rangle \mathbf{1} + t \sum_{\beta \in R_\Gamma^+} \langle v, \beta \rangle \langle w, \beta \rangle \frac{1 + q^\beta}{1 - q^\beta} \beta.$$

Bryan–Gholampour conjecture

If $\theta = \sum_i n_i \alpha_i$ is the highest root, then after analytic continuation to $q_i = \exp(2\pi i n_i / |\Gamma|)$ and the character-table linear change $y \mapsto x$, they conjecture that

$$QH^*(Y_\Gamma) \cong QH^*(\mathbb{C}^2/\Gamma).$$

Status: proved in type A by Coates–Corti–Iritani–Tseng and in type D by Hu.

One-dimensional Landau–Ginzburg mirror

Let W_q be a meromorphic function on Σ such that

$$\mathrm{Jac}(W) = \mathcal{O}_{u,q} / \frac{\partial W}{\partial u} \mathcal{O}_{u,q} \cong QH^*(Y),$$

then W_q is called a one-dimensional Landau–Ginzburg mirror of Y .

In particular, this means that one can identify

$$c_B(a, b, c) = \sum_{dW=0} \mathrm{Res} \frac{a b c}{dW} \phi^2, \quad \eta_B(a, b) = c(1, a, b),$$

with

$$c_A(a, b, c) = \eta_A(a * b, c), \quad \eta_A(a, b) = \int_Y a \cup b.$$

as isomorphism of Frobenius manifolds.

Landau-Ginzburg model from Hori-Vafa mirror

By Fang-Liu-Zong, the T' -equivariant Landau-Ginzburg mirror dimensionally reduces to the Hori-Vafa curve

$$C_q = \{H_q(X, Y) = 0\},$$

which carries the distinguished multivalued function

$$\hat{x} = u_1 x + u_2 y = -u_1 \log X - u_2 \log Y, \quad x = -\log X, \quad y = -\log Y.$$

Theorem: Toric Landau-Ginzburg mirror (Fang-Liu-Zong)

Under the toric mirror map, the residue Frobenius algebra of $\hat{x}$ is the equivariant quantum cohomology of X :

$$\text{Jac}(\hat{x}) \cong QH_{T'}^*(X).$$

Here u_1, u_2 are the equivariant parameters.

Geometric engineering

Nekrasov's gauge/integrable-system correspondence identifies the Seiberg–Witten curve of five-dimensional pure gauge theory on $\mathbb{R}^4 \times S^1$ with the affine relativistic Toda spectral curve. Geometric engineering realizes the same theory from the threefold obtained as the fiberwise Γ -quotient of the resolved conifold. Brini–Borot formulated this IS-SW-CY3 correspondence. Decompactifying the base $\mathbb{P}^1$ gives $[\mathbb{C}^2/\Gamma] \times \mathbb{C}$; the same limit gives the mirror curve relevant here.

Affine Toda curve

Let ρ be a fundamental representation of $\mathfrak{g}$, let $1 \leq \bar{k} \leq \text{rank } \mathfrak{g} = l$ be a marked index, and let χ_i be the i -th fundamental character. Then

$$\det_{\rho}(\mu - e^{\rho}) = (1 - \mu)^{n_0} Q(\mu, \chi_1, \dots, \chi_l).$$

The affine Toda curve is the zero locus of

$$P^{\Gamma}(\mu, \lambda) = Q(\mu, \chi_1, \dots, \chi_{\bar{k}} - \lambda/K, \dots, \chi_l).$$

Landau-Ginzburg model from affine Toda curve

The affine Toda curve $\Sigma_p = \{P^\Gamma(\mu, \lambda) = 0\}$ carries a distinguished meromorphic function

$$\lambda : \Sigma_p \longrightarrow \mathbb{P}^1.$$

Theorem: LG mirror symmetry for *ADE* resolutions (Brini–Ma–Strachan)

$$\mathrm{Jac}(\log \lambda) \cong QH^*(Y^\Gamma).$$

In fact, the ramification data of λ are independent of p , and the resulting Frobenius structure $(\log \lambda, \phi = d\mu/\mu)$ is Dubrovin's Hurwitz Frobenius manifold structure.

Taking the logarithm = applying Dubrovin duality

The orbit space of the extended affine Weyl group carries a Frobenius structure. Dubrovin's almost-duality replaces its product by

$$X \star Y = E^{-1} \circ X \circ Y,$$

where E is the Euler vector field.

On the Landau–Ginzburg side, this changes the superpotential from

$$\lambda \quad \text{to} \quad \log \lambda.$$

Remark

(λ, ϕ) is the mirror of $\mathbb{P}([\mathbb{C}^2/\Gamma])$.

Type A_n : Brini-Cavalieri-Ross

After choosing a marking, the affine A_n Toda curve can be written schematically as

$$\lambda(\mu) = K \mu^{-\bar{k}} \prod_{i=1}^{n+1} (\mu - \kappa_i).$$

Remark

At the orbifold point $\kappa_k = \exp(2\pi i \frac{k}{n+1})$, the curve is simply $\lambda = K \mu^{-\bar{k}} (\mu^{n+1} - 1)$, and the integrals over $[0, \kappa_k]$ reduce to Beta/Gamma integrals.

Type D_ℓ : superpotential with $\mathbb{Z}_2$ symmetry

Solving the spectral parameter in the D_ℓ case gives

$$\lambda \mu^{\ell-2} (\mu^2 - 1)^2 = K \prod_{i=1}^{\ell} (\mu - \kappa_i) (\mu - \kappa_i^{-1}).$$

There is an obvious involution $\iota : \mu \mapsto \mu^{-1}$ fixing λ . With $w = \mu + \mu^{-1}$, the Toda curve takes the form

$$\lambda(w) = K \frac{\prod_{i=1}^{\ell} (w - c_i)}{w^2 - 4}, \quad c_i = \kappa_i + \kappa_i^{-1}.$$

By the $\mathbb{Z}_2$ symmetry, the number of critical points doubles the the number of critical value, even for generic κ . Therefore one should consider the invariant part of $Jac(\log \lambda)$.

Givental–Teleman reconstruction

The Gromov–Witten classes naturally form a CohFT on $H^\bullet(X)$:

$$\Omega_{g,n}^X(\gamma_1, \dots, \gamma_n) = \sum_{\beta} q^{\beta} \operatorname{st}_* \left([\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\operatorname{vir}} \cap \prod_{i=1}^n \operatorname{ev}_i^*(\gamma_i) \right) \\ \in H^\bullet(\overline{\mathcal{M}}_{g,n}).$$

Theorem: Givental–Teleman reconstruction

A semisimple CohFT with unit is reconstructed from its genus-zero Frobenius structure and a calibrated R -matrix.

The calibrated matrix $R(x, z) = I + R_1(x)z + R_2(x)z^2 + \dots$, with $R(z)R(-z)^T = I$, satisfies the Dubrovin equation

$$z \, dS = C \, S, \quad \boxed{S(x, z) = \Psi(x) R(x, z) e^{U(x)/z}}.$$

The LG mirror theorem $\operatorname{Jac}(\log \lambda) \cong QH^*(Y^\Gamma)$ identifies C , U , and Ψ on the two sides.

R -matrix ambiguity

The Dubrovin equation leaves a diagonal ambiguity

$$\boxed{R_B = R_A D}, \quad D = \text{diag}(D_\alpha(z)),$$

Symplecticity further implies

$$D_\alpha(z) = \exp \left(\sum_{j \geq 0} \gamma_{\alpha,j} z^{2j+1} \right).$$

In order to extend to higher genus, one just need $D \equiv 1$.

B-model R -matrix from stationary-phase expansion

Put $W = -\hat{x}$, $u_\alpha = W(p_\alpha)$, and $U = \text{diag}(u_\alpha)$. Near p_α , choose a Morse coordinate with

$$W = u_\alpha - \frac{\zeta_\alpha^2}{2}.$$

Let ϑ_β^0 be the normalized primary second-kind differential for the canonical direction e_β : locally it has a double pole at p_β with zero residue. For the Lefschetz thimble Γ_α , set

$$\boxed{\hat{R}_{\beta\alpha}^B(z) \sim \frac{\sqrt{z} e^{-u_\alpha/z}}{2\sqrt{\pi}} \int_{\Gamma_\alpha}^{\text{reg}} e^{-\hat{x}/z} \vartheta_\beta^0} = \delta_{\beta\alpha} + \sum_{k \geq 1} (R_k^B)_{\beta\alpha} z^k.$$

Regularized stationary phase therefore determines every coefficient, giving

$$\hat{R}^B(z) = I + R_1^B z + R_2^B z^2 + \cdots, \quad S^B = \Psi^B \hat{R}^B(z) e^{U/z}.$$

Pairing with the unit isolates the diagonal gauge

Let $\{e_\beta\}$ be the normalized canonical basis, so that $\eta(e_\beta, e_\gamma) = \delta_{\beta\gamma}$. The unit has components

$$\mathbf{1} = \sum_{\beta} v_{\beta} e_{\beta}, \quad v_{\beta} = \eta(\mathbf{1}, e_{\beta}) = \frac{1}{\sqrt{\Delta_{\beta}}}.$$

Hence the unit component of the α -th column is

$$L_{\alpha}^B(z) := \eta(\mathbf{1}, R^B(z)e_{\alpha}) = \sum_{\beta} v_{\beta} R_{\beta\alpha}^B(z) = \frac{1}{\sqrt{-2}} \frac{e^{-u_{\alpha}/z}}{\sqrt{\pi z}} \int_{\Gamma_{\alpha}} e^{W/z} d\tilde{y}, \quad W = -\hat{x}.$$

Here $\phi_D/\mathfrak{g}_{\nu} = d\tilde{y}/\sqrt{-2}$ is the normalized primitive differential representing $\mathbf{1}$. Since $R^B = R^A D$ with $D = \text{diag}(D_{\alpha})$,

$$L_{\alpha}^B = D_{\alpha} L_{\alpha}^A.$$

Why one scalar period per column is enough

It does not reconstruct the full matrix; it determines the one remaining scalar gauge factor D_{α} .

Type A_n : the orbifold point fixes the normalization directly

In the Brini–Cavalieri–Ross calculation, the mirror curve is $\mathbb{P}^1$ with an especially simple superpotential, $\lambda = K\mu^{-\bar{k}}(\mu^{n+1} - 1)$.

- Lefschetz thimbles can be represented by $L_k = [0, \kappa_k]$.
- The oscillatory periods

$$\Pi_i(z) = \int_{L_i} \lambda^{1/z} \phi$$

become explicit Beta/Gamma integrals.

Theorem: Orbifold calibration (Brini–Cavalieri–Ross)

The stationary-phase asymptotics of Π_i give the same calibration as the A-side.

Type D_ℓ : degeneration at OP

Put $m = \ell - 2$ and $D_\sigma = \{\mu = \sigma\}$, $\sigma = \pm 1$. At the orbifold point, the central curve in the fixed surface is

$$(\mu^2 - 1)^2(\lambda\mu^m - K(\mu^{2m} + 1)) = 0,$$
$$C_0^{\text{flat}} = \Sigma^{\text{orb}} + 2D_+ + 2D_-, \quad \Sigma^{\text{orb}} : \lambda = K(\mu^m + \mu^{-m}).$$

Along a transverse parameter t , blow up the two double fibres and take the strict transform:

$$\mathfrak{S} = \text{Bl}_{(D_+ \cup D_-) \times \{0\}}(S \times \mathbb{A}_t^1), \quad \boxed{\bar{C}_0 = \Sigma^{\text{orb}} \cup \Sigma_+ \cup \Sigma_-}.$$

Thus the reduced central fibre is a tree: one main component and two exceptional bubbles $\Sigma_\pm$ attached at $\mu = \pm 1$.

Labels on the normalization at the orbifold point

On Σ^{orb} , $\mu^{2m} = 1$ gives $\ell - 3$ free ι -orbits plus two fixed main labels $\mu = \pm 1$. The bubbles contribute the two fixed labels $y_+ = y_- = 0$. Hence

$$\#I_{\text{orb}} = (\ell - 3) + 2 + 2 = \ell + 1.$$

Even and odd thimbles resolve the fixed labels

For each fixed node $r \in \{+1, -1\}$, let $p_{r,\varepsilon}$, $\varepsilon \in \{+, -\}$, be the two critical points on a nearby smooth fibre that collide at $\mu = r$. Let $\Gamma_{r,\varepsilon}$ be the Lefschetz thimble associated with $p_{r,\varepsilon}$. Define the even and odd combinations

$$\Gamma_r^{\text{even}} := \frac{\Gamma_{r,+} + \Gamma_{r,-}}{\sqrt{2}}, \quad \Gamma_r^{\text{odd}} := \frac{\Gamma_{r,+} - \Gamma_{r,-}}{\sqrt{2}}.$$

The local geometric label convention is

$$\Gamma_r^{\text{even}} \rightsquigarrow \sqrt{2} \Gamma_r^{\text{main}}, \quad \Gamma_r^{\text{odd}} \rightsquigarrow \sqrt{2} \Gamma_r^{\text{bub}}.$$

What the proof uses

The parity-even flat-unit limit fixes the calibration; the odd combination records the bubble label.

Boundary periods reproduce the A-model unit series

At the orbifold point, the A-model unit pairing is

$$L_{\alpha}^A(z) = \eta(\mathbf{1}, R_A(z)e_{\alpha}) = \eta(\mathbf{1}, e_{\alpha})S_{\text{unit}}(z) = v_{\alpha}(0)S_{\text{unit}}(z).$$

Here

$$S_{\text{unit}}(z) = \sqrt{\frac{\nu}{z}} \frac{\Gamma(\nu/z + 1/2)}{\Gamma(\nu/z + 1)}.$$

On the B-side, Euler's Beta integral evaluates the normalized main-component boundary period and reproduces the same unit pairing. Thus, for every free main label,

$$\lim_{\kappa \rightarrow \kappa^{\text{orb}}} L_{\alpha}^B(\kappa; z) = v_{\alpha}(0)S_{\text{unit}}(z).$$

Theorem: Free-label gauge fixing

Since $L_{\alpha}^B = D_{\alpha}L_{\alpha}^A$, matching the two unit pairings forces $D_{\alpha} = 1$ for every free label.

Parity-even boundary data fix the remaining labels

For each pair of smooth labels approaching a fixed node $r = \pm 1$ along the chosen boundary arc,

$$\lim_{\kappa \rightarrow \kappa^{\text{orb}}} \frac{L_{r,+}^B + L_{r,-}^B}{\sqrt{2}} = \sqrt{2} v_r S_{\text{unit}}(z).$$

Comparison with the A-model gives

$$D_+(z) + D_-(z) = 2.$$

Symplecticity also gives

$$D_{\pm}(z)D_{\pm}(-z) = 1.$$

Together these imply $D_+ = D_- = 1$.

$R_B = R_A.$

Matching the R -matrix calibrations

Theorem: Calibration match

On the smooth oscillatory chamber, in the normalized canonical frame,

$$R_B(\kappa; z) = R_A(x(\kappa); z).$$

The orbifold point supplies coefficientwise boundary values. Since its central fibre is singular and non-semisimple, no central-fibre B-model R -matrix is defined. Fixing the base-constant gauge at the boundary and analytically continuing through the chamber gives

$$\text{GW} = \text{Givental}(R_A) = \text{Givental}(R_B).$$

The remaining bridge

To complete the mirror theorem, it remains to translate Givental–Teleman reconstruction into topological recursion, using the correspondence of Dunin-Barkowski, Orantin, Shadrin, and Spitz (DOSS).

Givental formula = topological recursion

A DOSS-type theorem rewrites the topological-recursion invariants as a Givental graph sum:

$$\mathrm{TR}(\mathcal{S}) = \mathrm{Givental}(R_B).$$

The local spectral-curve data determine

- the canonical TFT at the vertices;
- the R_B -propagator on the edges;
- the dilaton and ordinary leaves.

$$\mathrm{GW} = \mathrm{Givental}(R_A) = \mathrm{Givental}(R_B) = \mathrm{TR}.$$

The B-model of binary-dihedral Calabi–Yau threefolds

Let Γ be a binary dihedral group and

$$\mathcal{X} = [\mathbb{C}^2/\Gamma \times \mathbb{C}].$$

This is the affine Calabi–Yau threefold associated with the type- D_ℓ McKay data.

A-model

The equivariant orbifold GW theory of $\mathcal{X}$.

B-model

The corresponding topological recursion on the type- D_ℓ Toda curve.

The involution and $\mathbb{Z}_2$ -equivariant topological recursion of Giacchetto–Kramer–Lewński

The involution

$$\iota : \mu \longmapsto \mu^{-1}$$

fixes the superpotential but reverses the primitive differential. Ordinary recursion upstairs sees twice the desired canonical state space.

The sign-sector specialization of the $\mathbb{Z}_2$ -equivariant topological recursion of Giacchetto–Kramer–Lewński uses the Prym kernel

$$B^{\text{std}}(\mu_1, \mu_2) = \frac{d\mu_1 d\mu_2}{(\mu_1 - \mu_2)^2}, \quad B^- = B^{\text{std}} - (\text{id} \times \iota)^* B^{\text{std}}.$$

Its correlators are anti-invariant in every external variable.

Equivariant recursion and descendant inputs

Set

$$\mathcal{S}_\kappa^\iota = (\Sigma_\Gamma, x, y, B^-).$$

The Giacchetto–Kramer–Lewński $\mathbb{Z}_2$ -equivariant topological recursion in the sign sector defines

$$\omega_{g,n}^\iota := \mathrm{TR}_{g,n}^{\mathbb{Z}_2}(\mathcal{S}_\kappa^\iota), \quad 2g - 2 + n > 0,$$

with $\iota_j^* \omega_{g,n}^\iota = -\omega_{g,n}^\iota$ in every variable.

Parity-twisted B-model descendant series

Let $\mathcal{W}_{g,n}^{B,\Pi}$ be the series obtained by expanding the equivariant-recursion correlators in the canonical external-leaf basis and replacing each leaf by the corresponding A-model descendant variable.

The superscript Π records the parity signs in this replacement.

Closed remodeling for the binary-dihedral threefold

On the smooth chamber, the mirror map gives:

Theorem: Stable descendants: $n > 0, 2g - 2 + n > 0$

$$\mathcal{W}_{g,n}^{B,\Pi}(\tilde{\mathbf{u}}_1, \dots, \tilde{\mathbf{u}}_n; \kappa) = \mathcal{F}_{g,n}^{\mathcal{X},\text{an}}(\mathbf{u}_1, \dots, \mathbf{u}_n; \kappa).$$

Theorem: Free energies: $g \geq 2$

$$F_g(\mathcal{S}_\kappa^\iota) = F_g^{\text{GW},\text{an}}(\mathcal{X}; \kappa).$$

The ADE construction extends the toric remodeling paradigm

	Toric / FLZ	ADE / Toda
Input	toric diagram	ADE root system
Mirror	Hori–Vafa curve	affine Toda spectral curve
Genus zero	toric mirror theorem	ADE LG mirror theorem
Calibration	standard toric boundary data	semistable orbifold degeneration in type D
Recursion	ordinary EO recursion	$\mathbb{Z}_2$ equivariant recursion

Takeaway

Affine relativistic Toda systems provide natural one-dimensional Landau–Ginzburg mirrors for ADE resolutions.

The genus-zero theorem identifies these mirrors with ADE quantum cohomology.

In type D , a semistable orbifold degeneration fixes the calibration, and DOSS turns the same curve into a non-toric remodeling theorem.

In one line

ADE root system $\Rightarrow$ Toda LG mirror $\Rightarrow$ calibrated R -matrix $\Rightarrow$ topological recursion.

Thank you

Questions?